

Asset Master

1. Metadata

Field	Value
Version	1.0
Owner	Usha
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OverView

Service: FAM.API

Module: AssetMasterGeneral

The **purpose of the** AssetMasterGeneral

AssetMasterGeneral is the system-of-record for fixed assets. It stores identity (code, name), classification (group/sub-group/category), structure (parent/child), technical flags (asset type, working status, tangible/depreciable), quantities/UOM, documents & image paths, purchase/warranty/insurance links, and operational relationships (location, transfers, disposal).

Out of scope

- Depreciation calculation (handled by depreciation engine)
- Accounting postings (Finance)
- File storage service (this module only stores paths/URIs)

Goals

- Single source of truth for asset identity & hierarchy
- Strong referential integrity to master data (Group/SubGroup/Category/Misc/UOM)
- Auditable create/update/delete (soft delete)
- Rich read APIs (list, detail, split view, autocomplete)
- Utility APIs (code pattern, parent lookup, types, working status)

- File operations (logo/document upload/delete), Excel imports, physical audit & scanning

2. Data Model

```
1 public class AssetMasterGenerals : BaseEntity
2 {
3     public int CompanyId { get; set; }
4     public int UnitId { get; set; }
5     public string? AssetCode { get; set; }
6     public string? AssetName { get; set; }
7
8     public int AssetGroupId { get; set; }
9     public AssetGroup AssetGroup { get; set; } = null!;
10    public int? AssetSubGroupId { get; set; }
11    public AssetSubGroup? AssetSubGroup { get; set; } = null!;
12    public int AssetCategoryId { get; set; }
13    public AssetCategories AssetCategories { get; set; } = null!;
14    public int AssetSubCategoryId { get; set; }
15    public AssetSubCategories AssetSubCategories { get; set; } = null!;
16
17    public int? AssetParentId { get; set; }
18    public AssetMasterGenerals AssetParent { get; set; } = null!;
19    public ICollection<AssetMasterGenerals> AssetChildren { get; set; }
20
21    public int? AssetType { get; set; }
22    public MiscMaster? AssetMiscType { get; set; } = null!;
23    public string? MachineCode { get; set; }
24    public int? Quantity { get; set; }
25    public int? UOMId { get; set; }
26    public UOM UomMaster { get; set; } = null!;
27    public string? AssetDescription { get; set; }
28
29    public int? WorkingStatus { get; set; }
30    public MiscMaster? AssetWorkType { get; set; } = null!;
31    public string? AssetImage { get; set; }
32    public byte ISDepreciated { get; set; }
33    public byte IsTangible { get; set; }
34    public string? AssetDocument { get; set; }
35    public DateTimeOffset? PutToUseDate { get; set; }
36
37    // Relations (read-only here)
38    public ICollection<AssetPurchaseDetails>? AssetPurchase { get; set; }
39    public ICollection<AssetSpecifications>? AssetSpecification { get; set; }
40    public ICollection<AssetWarranties>? AssetWarranty { get; set; }
41    public ICollection<AssetAdditionalCost>? AssetAdditionalCost { get; set; }
42    public AssetLocation? AssetLocation { get; set; }
43    public ICollection<AssetAmc>? AssetAmc { get; set; }
44    public ICollection<AssetInsurance>? AssetInsurance { get; set; }
45    public AssetDisposal? AssetDisposalMaster { get; set; }
46    public ICollection<DepreciationDetails>? DepreciationDetails { get; set; }
47    public ICollection<AssetTransferIssueDtl>? AssetTransferIssueMaster { get; set; }
48    public ICollection<AssetTransferReceiptDtl>? AssetTransferReceiptMaster { get;
49    set; }
50 }
```

Column	Type	Constraints / Notes
Id	int	PK , identity
CompanyId	int	Required
UnitId	int	Required
AssetCode	varchar(50)	Required, Unique within (CompanyId, UnitId) (recommended)
AssetName	varchar(100)	Required
AssetGroupId	int	Required , FK → AssetGroup (Restrict)
AssetSubGroupId	int	Nullable, FK → AssetSubGroup (Restrict)
AssetCategoryId	int	Required , FK → AssetCategories (Restrict)
AssetSubCategoryId	int	Required , FK → AssetSubCategories (Restrict)
AssetParentId	int	Nullable self-FK (Restrict, prevents cyclic delete)
AssetType	int	Nullable, FK → MiscMaster (“Asset Type”, Restrict)
MachineCode	varchar(50)	Nullable
Quantity	int	Nullable (≥1 when provided)
UOMId	int	Nullable, FK → UOM (Restrict)

AssetDescription	varchar(250)	Nullable
WorkingStatus	int	Nullable, FK → MiscMaster ("Working Status", Restrict)
AssetImage	nvarchar(255)	Nullable (URI/path)
AssetDocument	nvarchar(255)	Nullable (URI/path)
ISDepreciated	bit	Required (stored as bit; expose bool)
IsTangible	bit	Required (stored as bit; expose bool)
PutToUseDate	datetimeoffset	Nullable
IsActive	bit	Required , default 1
IsDeleted	bit	Required , default 0
Audit fields	—	Required on create

- Table FixedAsset.AssetMaster
- Enum → bit converters for IsActive , IsDeleted
- Byte ↔ bit converters for IsTangible , ISDepreciated
- All FKs Restrict (no cascade) to protect history

3. API Contracts

#	Endpoint	Method	Purpose
1	/api/AssetMasterGeneral	GET	Paged list with optional SearchTerm
2	/api/AssetMasterGeneral	POST	Create (CQRS: CreateAssetMasterGeneralCommand)

3	<code>/api/AssetMasterGeneral</code>	PUT	Update (CQRS: <code>UpdateAssetMasterGeneralCommand</code>)
4	<code>/api/AssetMasterGeneral/{id}</code>	GET	Get asset by id
5	<code>/api/AssetMasterGeneral/{id}</code>	DELETE	Soft delete (CQRS: <code>DeleteAssetMasterGeneralCommand</code>)
6	<code>/api/AssetMasterGeneral/{id}/split</code>	GET	Split view (returns grouped tabs/sections DTO)
7	<code>/api/AssetMasterGeneral/by-name?name=</code>	GET	Autocomplete by name/code
8	<code>/api/AssetMasterGeneral/AssetType</code>	GET	List Misc “Asset Type”
9	<code>/api/AssetMasterGeneral/ParentAsset?assetType=</code>	GET	Find possible parents by <code>assetType</code>
10	<code>/api/AssetMasterGeneral/WorkingStatus</code>	GET	List Misc “Working Status”

11	<code>/api/AssetMasterGeneral/AssetCodePattern</code>	GET	Get current asset code pattern/config
12	<code>/api/AssetMasterGeneral/upload-logo</code>	POST	Upload/replace asset image/logo (returns path)
13	<code>/api/AssetMasterGeneral/delete-logo</code>	DELETE	Delete asset image/logo (by path payload)
14	<code>/api/AssetMasterGeneral/import</code>	POST	Excel import (master data)
15	<code>/api/AssetMasterGeneral/upload-document</code>	POST	Upload attachment, returns path
16	<code>/api/AssetMasterGeneral/delete-document</code>	DELETE	Delete attachment (by path payload)
17	<code>/api/AssetMasterGeneral/saveDocument</code>	POST	Persist document metadata to asset
18	<code>/api/AssetMasterGeneral/PhysicalAuditUpload</code>	POST	Excel import for physical audit (≤ 2 MB)
19	<code>/api/AssetMasterGeneral/Sc</code>	POST	Scan/verify an asset (barcode/Rfid)

4. Validation & Business Rules

4.1 Validation (FluentValidation)

CreateDepGroupCommand

- **CompanyId, UnitId**: required; must exist (tenant scope).
- **AssetCode**: required, `<= 50`, trimmed/upper; **unique within (CompanyId, UnitId)**.
- **AssetName**: required, `<= 100`.
- **AssetGroupId / AssetCategoryId / AssetSubCategoryId**: required; must exist & not deleted; pair (Category ↔ Group) should be consistent (repository check).
- **AssetSubGroupId**: optional; if provided, must belong to **same AssetGroup**.
- **AssetParentId**: optional; if provided, must exist **in same Company/Unit**, and **cannot create cycles**.
- **AssetType / WorkingStatus**: if provided, must exist in **MiscMaster** with the proper category.
- **Quantity**: if provided, `>= 1`.
- **UOMId**: if provided, must exist in UOM master.
- **IsTangible / ISDepreciated**: boolean in API; map to byte/bit; business rule: intangible ⇒ `ISDepreciated = false` (enforce).
- **PutToUseDate**: must be `<=.UtcNow + tolerance` and `>= purchase date` when available.

5.2 Delete (soft)

- Set `IsDeleted = 1`; exclude from default reads.
- **Block** delete when active dependencies exist (policy):
 - has non-deleted depreciation rows, transfers in progress, open AMC/insurance, or is parent of active children.
 - Otherwise require deactivation (`IsActive = 0`) first.

5.3 Additional rules

- **Hierarchy**: parent/child allowed only within same Company/Unit.

- **Code pattern:** creation **should** comply with `/AssetCodePattern` output (validate if setting enabled).
- **Audit:** fill `Created*` / `Modified*` from user/IP services.
- **Events/Outbox:** publish `AssetCreated/Updated/Deleted` , `AssetDocumentAdded/Removed` , etc.
- **Ordering:** default list order by `AssetCode` , `AssetName` .
- **Security:** enforce RBAC; no PII in logs; sanitize file names.

5. Error Handling & Status Codes Error Handling & Status Codes

- **200 OK** – Successful read/update/delete.
- **201 Created** – Successful create.
- **400 Bad Request** – Validation failed/invalid parameters.
- **404 Not Found** – Resource not found.
- **409 Conflict** – Uniqueness violations

6. Dependencies

- **MediatR** for CQRS handlers (`GetAssetMasterGeneralQuery` , `CreateAssetMasterGeneralCommand` , etc.)
- **FluentValidation** for request validation.
- **EF Core / Dapper** for persistence (implementation-specific).
- **Serilog** for structured logging.